MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, GERMANIUM, LOW-POWER TYPE 2N358A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a NPN, germanium, low-power transistor.

1.2 Physical dimensions. See figure 1 (TO-5).

1.3 Maximum ratings.

P_T 1/	V_{CBO}	V_{EBO}	V_{RT}	T_{stg}
mW	Vdc	Vdc	Vdc	°C
150	30	20	25	-65 to +100

1/ Derate linearly 2 mW/°C for $T_A > 25°C$.

1.4 Primary electrical characteristics.

Limits	h_{FE} V_{CE} = 0.25 Vdc I_C = 300 mAdc	h_{FE} V_{CE} = 0.25 Vdc I_B = 1 mAdc	V_{CE}(sat) I_C = 300 mAdc I_B = 30 mAdc	V_{BE} V_{CE} = 0.25 Vdc I_C = 300 mAdc
			Vdc	Vdc
Min	25	65	---	---
Max	75	---	0.2	1.25

2. APPLICABLE DOCUMENTS

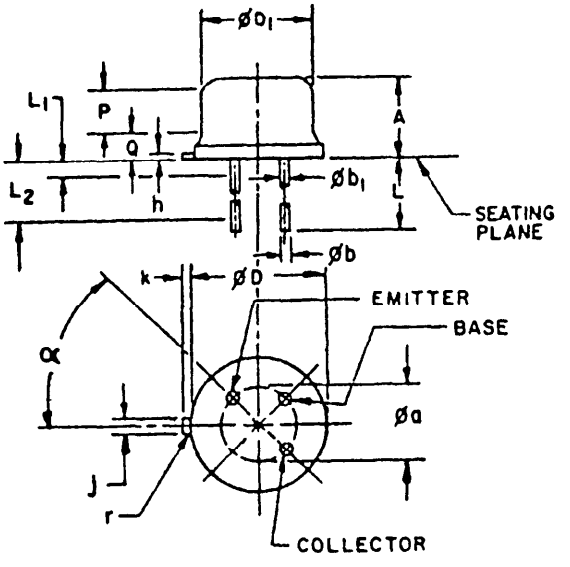
2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

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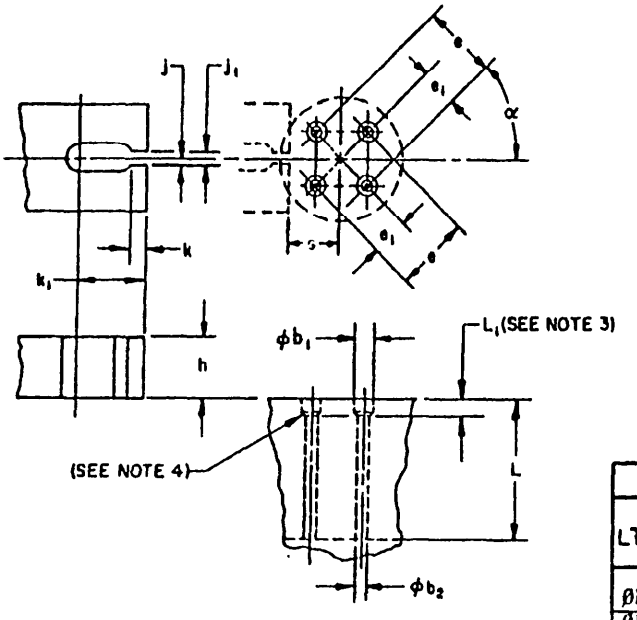


LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.240	.260	6.10	6.60	
øa	.200 TP		5.08 TP		6
øb	.016	.021	.41	.53	7, 8
$øb_1$	.016	.019	.41	.48	7, 8
øD	.335	.370	8.51	9.40	
$øD_1$	.305	.335	7.75	8.51	
h	.009	.041	.23	1.04	
j	.028	.034	.71	.86	2
k	.029	.045	.74	1.14	3
L	1.500	1.750	38.10	44.45	7, 8
L_1	---	.050	---	1.27	7, 8
L_2	.250	---	6.35	---	7, 8
P	.100	---	2.54	---	5
Q	---	.050	---	1.27	4
r	---	.010	---	.25	10
α	45° TP		45° TP		6

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Beyond r(radius) maximum, j shall be held for a minimum length of .011(.28 mm).
3. k measured from maximum øD.
4. Outline in this zone is not controlled.
5. $øD_1$ shall not vary more than .010(.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gage plane .054 +.001-.000(1.37+.03-.00 mm) below seating plane shall be within .007(.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gage and gaging procedure shown in figure 2.
7. $øb_1$ applies between L_1 and L_2. øb applies between L_2 and L minimum. Diameter is uncontrolled in L_1 and beyond L minimum.
8. All three leads.
9. The base lead is electrically connected to the case.
10. r(radius) applies to both inside corners of tab.

FIGURE 1. Physical dimensions of transistor type 2N358A (TO-5).

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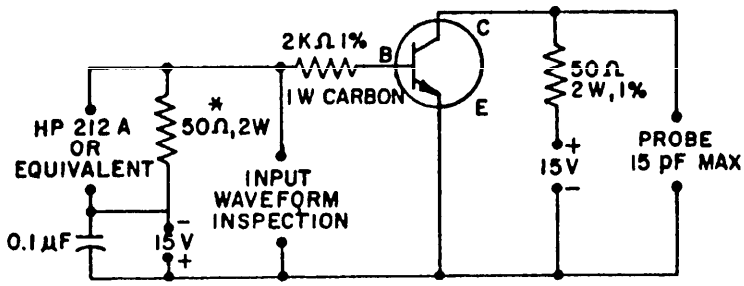


LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
$øb_1$	.0595	.0605	1.51	1.54	
$øb_2$	.0325	.0335	.83	.85	
e	.1995	.2005	5.07	5.09	
e_1	.0995	.1005	2.53	2.55	
h	.150 Nominal		3.81 Nominal		
j	.0175	.0180	.44	.46	
j1	.0350	.0355	.89	.90	
k	.009	.011	.23	.28	
k_1	.125 Nominal		3.18 Nominal		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
S	.182	.199	4.62	5.05	1
α	44.90°	45.10°	44.90°	45.10°	

NOTES:
1. The location of the tab locator within the limits indicated will be determined by the tab and flange dimensions of the device being checked.
2. Gaging procedure. The device being measured shall be inserted until its seating plane is .125 ± .010 (3.18 ± .25 mm) from the seating surface of the gage. A force of 8 ± .5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gage. The use of a pin straightener prior to insertion in the gage is permissible.
3. Gaging plane.
4. Drill Angle.

FIGURE 2. Gage for lead and tab location for transistor type 2N358A.

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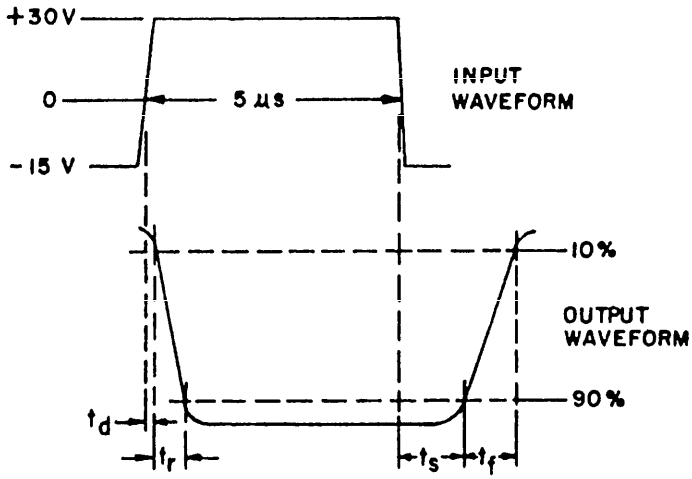


FIGURE 3. Pulse response test circuit.

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STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Transistors shall be of the design, construction, and physical dimensions shown on figure 1.

3.3.1 Lead material and finish. Lead material shall be Kovar or alloy 52. Lead finish shall be gold-plated. (Leads may be tin-coated if specified in the contract or order, and this requirement shall not be construed as adversely affecting the qualified-product status of the device, or applicable JAN marking, see 6.2).

3.3.1.1 Selectivity of lead material. Where choice of lead material (see 3.3.1 above) is desired, it shall be specified in the contract or order (see 6.2).

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A, B, and C inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall consist of the tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every 6 months during production.

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TABLE I. Group A inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Breakdown voltage, collector to emitter	3011	Bias condition D I_C = 600 μAdc		BV_{CEO}	15	---	Vdc
Collector to base cutoff current	3036	Bias condition D V_{CB} = 30 Vdc		I_{CBO}	---	25	μAdc
Collector to base cutoff current	3036	Bias condition D V_{CB} = 5 Vdc		I_{CBO}	---	5	μAdc
Collector to emitter cutoff current	3041	Test condition A; V_{CE} = 20 Vdc; V_{BB} = -1.5 Vdc; R_{BE} = 62 kilohms		I_{CEX}	---	10	μAdc
Emitter to base cutoff current	3061	Bias condition D V_{EB} = 20 Vdc		I_{EBO}	---	25	μAdc
Emitter to base cutoff current	3061	Bias condition D V_{EB} = 5 Vdc		I_{EBO}	---	5	μAdc
Reach through voltage	---	V_{EB} = 1 Vdc (see 4.4.2)		V_{RT}	25	---	Vdc
Subgroup 3			5				
Forward-current transfer ratio	3076	V_{CE} = 0.25 Vdc; I_C = 300 mAdc pulsed (see 4.4.1)		h_{FE}	25	75	---
Forward-current transfer ratio	3076	V_{CE} = 0.25 Vdc; I_B = 1 mAdc		h_{FE}	65	---	---
Base emitter voltage (nonsaturated)	3066	Test condition B V_{CE} = 0.25 Vdc; I_C = 300 mAdc pulsed (see 4.4.1)		V_{BE}	---	1.25	Vdc
Base emitter voltage (nonsaturated)	3066	Test condition B V_{CE} = 0.25 Vdc; I_B = 1 mAdc		V_{BE}	---	0.4	Vdc
Collector to emitter voltage (saturated)	3071	I_C = 300 mAdc; I_B = 30 mAdc pulsed (see 4.4.1)		$V_{CE}(sat)$	---	0.2	Vdc
Subgroup 4			10				
Turn-on time	---	See figure 3		t_{on}	---	0.8	μs
Storage time	---	See figure 3		t_s	---	0.6	μs
Fall time	---	See figure 3		t_f	---	0.8	μs

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TABLE I. Group A inspection. - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 6							
High-temperature operation:		T_A = +55°C					
Collector to base cutoff current	3036	Bias condition D V_{CB} = 15 Vdc		I_{CBO}	---	50	µAdc
Low-temperature operation:		T_A = -65°C					
Forward-current transfer ratio	3076	V_{CE} = 1 Vdc I_C = 10 mAdc		h_{FE}	30	---	---

TABLE II. Group B inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Physical dimensions	2066	See figure 1		---	---	---	---
Subgroup 2			15				
Solderability	2026			---	---	---	---
Thermal shock (temperature cycling)	1051	Test condition B, except in step 3, T_A = +100°C +5, -0°C		---	---	---	---
Thermal shock (glass strain)	1056	Test condition A		---	---	---	---
Hermetic seal	1071	Test condition G or H for fine leaks; test condition A, C, D or F for gross leaks		---	---	$1x10^{-7}$	atm cc/s
Moisture resistance	1021			---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias condition D V_{CB} = 30 Vdc		I_{CBO}	---	25	µAdc
Forward-current transfer ratio	3076	V_{CE} = 0.25 Vdc I_C = 300 mAdc pulsed (see 4.4.1)		h_{FE}	25	75	---
Subgroup 3			15				
Shock	2016	Nonoperating; 500 G; 1 ms; 5 blows in each orientation: X_1, Y_1, Y_2 and Z_1		---	---	---	---
Vibration, variable frequency	2056			---	---	---	---

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TABLE II. Group B inspection. - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 3 - Continued							
Constant acceleration	2006	10,000 G; in each orientation: X_1, Y_1, Y_2 and Z_1		---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 4			20				
Terminal strength (lead fatigue)	2036	Test condition E		---	---	---	---
End points:							
Hermetic seal	1071	Test condition G or H for fine leaks; test condition A, C, D or F for gross leaks		---	---	$1x10^{-7}$	atm cc/s
Subgroup 5			20				
Salt atmosphere (corrosion)	1041	Electrical rejects may be used		---	---	---	---
Subgroup 6			7				
High-temperature life (nonoperating)	1032	T_{stg} = +100°C (see 4.3.4)		---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias condition D V_{CB} = 30 Vdc		I_{CBO}	---	35	μAdc
Forward-current transfer ratio	3076	V_{CE} = 0.25 Vdc I_C = 300 mAdc pulsed (see 4.4.1)		h_{FE}	20	90	---
Subgroup 7			7				
Steady-state operation life	1027	P_T = 150 mW V_{CB} = 15 Vdc (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6)							

TABLE III. Group C inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Resistance to solvents	---	MIL-STD-202, Method 215 (see 4.4.3)		---	---	---	---

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TABLE III. Group C inspection. - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 2			λ = 10				
High-temperature life (nonoperating)	1031	T_{stg} = +100°C (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6 of group B)							
Subgroup 3			λ = 10				
Steady-state operation life	1026	P_T = 150 mW; V_{CB} = 15 Vdc (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6 of group B)							

4.3.4 Group B and group C life-test samples. Samples that have been subjected to group B, 340-hour life-test, may be continued on test to 1,000 hours in order to satisfy group C life-test requirements. These samples shall be predesignated, and shall remain subjected to the group C, 1,000-hour acceptance evaluation after they have passed the group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1,000 hours shall be computed for 1,000-hour acceptance criteria (see 4.3.3).

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, and as follows:

4.4.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.4.2 Voltage, reach-through, collector to emitter. An adequate forward dc current shall be applied to the emitter terminal, and a reverse dc voltage shall be applied between collector and base, starting at zero and increasing until the emitter to base voltage has reached the specified reverse value. The dc voltage between the collector and the base terminals shall then be measured. The V_{RT} will be equal to the reverse collector to base dc voltage less the reverse emitter to base dc voltage.

4.4.3 Resistance to solvents. Transistors shall be subjected to tests in accordance with method 215 of MIL-STD-202. The following details shall apply:

(a) All areas of the transistor body where marking has been applied shall be brushed.
(b) After subjection to the tests, there shall be no evidence of mechanical damage to the device and markings shall have remained legible.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data.

(a) Lead finish if other than gold-plated (see 3.3.1).
(b) Selectivity of lead material (see 3.3.1.1).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Navy - EC
NASA - NA
Air Force - 17

Review activities:
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Air Force - 11, 80
DSA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS, SH
Air Force - 13, 15, 19

Preparing activity:
Navy - EC

Agent:
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